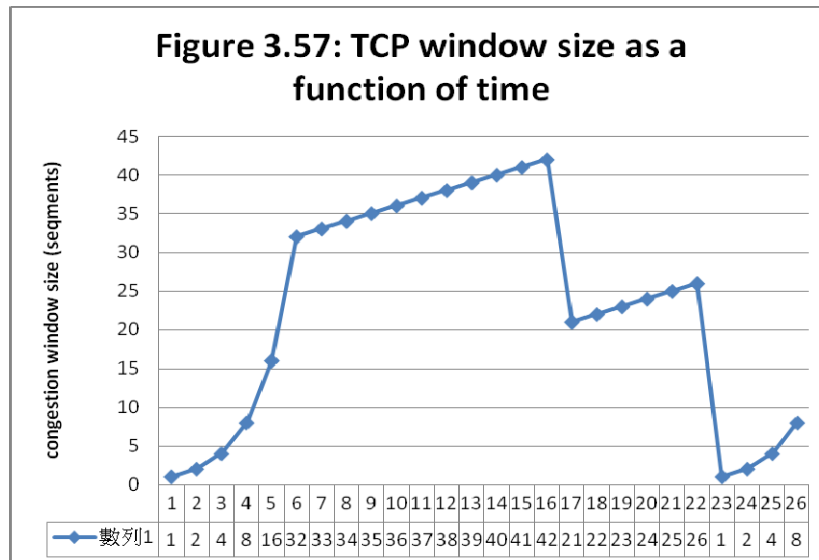


Chapter 3

Problems:

P32.

Consider the following plot of TCP window size as a function of time.



Assuming TCP Reno is the protocol experiencing the behavior shown above, answer the following questions. In all cases, you should provide a short discussion justifying your answer.

- What is the value of Threshold at the 18th transmission round?
- What is the value of Threshold at the 24th transmission round?
- Identify the intervals of time when TCP slow start is operating.
- Assuming a packet loss is detected after the 26th round by the receipt of a triple duplicate ACK, what will be the values of the congestion window size and Threshold?
- After the 16th transmission round, is segment loss detected by a triple duplicate ACK or by a timeout?
- After the 22nd transmission round, is segment loss detected by a triple duplicate ACK or by a timeout?
- Identify the intervals of time when TCP congestion avoidance is operating.
- What is the initial value of Threshold at the first transmission round?
- During what transmission round is the 70th segment sent?

P31. Recall the idealized model for the steady-state dynamics of TCP. In the period of time from when the connection's window size varies from $\frac{W}{2}$ (MSS) to W (MSS), only one packet is lost (at the very end of the period).

(a.) show that the loss rate is equal to $L = \text{loss rate} =$

$$\frac{1}{\left(\frac{3}{8}w^2 + \frac{3}{4}w\right)}$$

(b.) Use the result above to show that if a connection has loss rate L , then its average bandwidth is approximately given by

$$\approx \frac{1.22 * MSS}{RTT \sqrt{L}}$$